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Perfect Competition

Assumptions of a Perfectly Competitive Market

A perfectly competitive market is defined by a specific set of structural assumptions, and the Juice Mogul simulation was built to enforce every one of them. First, the market contains many buyers and many sellers, each too small individually to influence the market price; with twelve juice firms competing for the same customer base, no single seller could move the price on its own. Second, every firm sells an identical, undifferentiated product, so buyers have no reason to prefer one seller's juice over another's except on price. Third, firms and consumers have perfect information about prices, meaning a buyer can instantly see and compare what every seller charges. Fourth, entry into and exit from the market is free, so firms can adjust production without facing legal or financial barriers. Finally, because of these conditions, every firm is a price taker: it must accept the prevailing market price rather than set its own, and its only real decision is how much output to produce at that price (Shapiro et al., 2022).

Game Strategy and Results

My strategy going into the simulation was to test the boundaries of the market rather than simply accept the going price, and I alternated between pricing above and below what the rest of the market was charging. Pricing above the market price backfired immediately: because the juice sold in the game is a homogeneous good and buyers have perfect information, customers had no incentive to pay more for my product when eleven identical alternatives were available at a lower price, so my sales collapsed toward zero while fixed costs kept accruing. Pricing below the market price had the opposite problem — it did generate sales volume, but each unit was sold for less than it cost to produce, so higher volume simply multiplied the losses rather than building profit. By the end of the simulation, my cumulative profit was -$9,640.90, placing me last on the twelve-firm leaderboard. Every one of the eleven AI-controlled competitors — Healthy Dose, Vitafusion, Fresh Fusion, Fruit & Veggie Bar, Just Juice, Just Squeezed, Smoothie-Licious, The Juice Caboose, The Juice Press, Oasis Juice Bar, and Juice Do

It — finished with an identical profit of $2,500.00 (see Appendix), which strongly suggests that the profit-maximizing move in this environment was simply to accept the market price and produce the quantity at which price equals marginal cost, rather than to experiment with it.

Is This Market Efficient?

The uniform $2,500 profit earned by every competitor other than me is itself evidence of an efficient market at work. In perfect competition, free entry and exit push economic profit toward zero in the long run, so the $2,500 each firm earned likely represents a normal return built into the cost structure rather than true economic profit — exactly what the textbook model predicts (Shapiro et al., 2022). The market is efficient in two respects. Productively, competition forces firms to produce at their lowest average cost, since any firm producing less efficiently would be undercut by rivals selling the identical good for less. Allocatively, price settles where it equals marginal cost, meaning resources are used to produce exactly the quantity consumers value at that cost — no more, no less. The forces driving this efficiency are the same assumptions discussed above: perfect information eliminates price dispersion since buyers can instantly find the lowest price, product homogeneity removes any non-price basis for competition, and free entry and exit ensures that any firm pricing above marginal cost is punished by losing all its customers. My own result does not represent a market inefficiency; it represents the market efficiently disciplining a firm — mine — that deviated from the competitive price rather than accepting it.

What I Learned

The clearest lesson from this simulation is what it actually means to be a price taker. Coming in, I assumed there would be some strategic advantage to be found in pricing decisions, the way there might be in a monopoly or oligopoly setting. Perfect competition eliminated that advantage entirely: because the product was homogeneous and information was perfect, any attempt to charge more than the market price was punished instantly and completely, and any attempt to charge less simply converted volume into deeper losses. The only lever available to a firm in this market structure is the output decision — how much to produce at the price the market has already set — and the eleven AI firms that stuck to that lever outperformed my more aggressive pricing experiments by a wide margin.

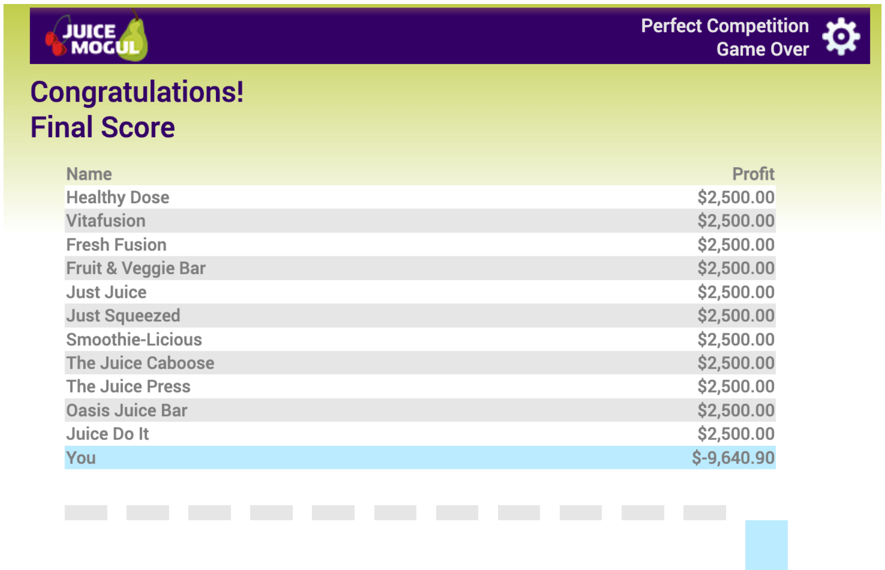
What I Would Do Differently

If I played the simulation again, I would stop treating price as a variable to be tested and instead treat it as a given, focusing my decisions entirely on quantity. Before setting output, I would estimate my marginal and average variable cost at each production level and choose the quantity where marginal cost equals the market price, shutting down production altogether if the market price fell below my average variable cost. I would also make smaller, more deliberate adjustments round to round based on cost data rather than large experimental swings in price, which is closer to how the eleven firms that each earned $2,500 appear to have played the game.

Appendix

Figure 1

Final Leaderboard, Juice Mogul: Perfect Competition Simulation



Name	Profit
Healthy Dose	$2,500.00
Vitafusion	$2,500.00
Fresh Fusion	$2,500.00
Fruit & Veggie Bar	$2,500.00
Just Juice	$2,500.00
Just Squeezed	$2,500.00
Smoothie-Licious	$2,500.00
The Juice Caboose	$2,500.00
The Juice Press	$2,500.00
Oasis Juice Bar	$2,500.00
Juice Do It	$2,500.00
You	$-9,640.90

Note. Screenshot of the final profit standings from the Juice Mogul: Perfect Competition simulation, showing a cumulative loss of -$9,640.90 compared with the uniform $2,500 profit earned by each of the eleven competing firms.